MODULE 09 · DOMAIN 2 - CLAUDE MODELS, PROMPTING AND CONTEXT ENGINEERING

Context Windows and Token Budgets

Why more context is not better, token tactics at each stage, and keeping long conversations and agents on track.

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A bigger window is capacity, not a strategy. Every token you add **competes for attention**.

Context rot

- **Recall falls as context grows.** Anthropic calls this context rot.
- **Current models hold up to 1M tokens**, and up to 600 images or PDF pages.
- **Cost and latency rise with input,** on every single request.
- **Irrelevant material is not neutral:** it dilutes what matters.

> ON THE EXAM
>
> Aim for the smallest set of high-signal tokens, not the largest window you can fill.

Token tactics, by stage of the request

TACTIC	STAGE
Summarise earlier conversation history	Input preparation
Trim retrieved passages to the relevant spans	Input preparation
Put stable content first, as a cacheable prefix	Prompt construction
Long documents near the top, question at the end	Prompt construction
Set max tokens to fit the expected output	Output handling
Validate the output against a schema	Output handling
Persist results so they are never regenerated	Output handling

ON THE EXAM

Input preparation shrinks what goes in; construction orders it; output handling bounds and keeps what comes out.

Extract or summarise?

EXTRACT THE SPANS

Exact words kept

- Quotes and citations survive
- Figures stay exact
- Best for evidence

SUMMARISE

The gist kept

- Compresses long history
- Detail is lost
- Best for continuity

THE TIEBREAKER

Need the exact words, figures or citations? Extract. Need the gist of what happened? Summarise.

Keeping long sessions on track

TOOL	WHAT IT DOES
Compaction	Summarises earlier context once a threshold is passed
Context editing	Clears old tool results and thinking blocks
Memory tool	Claude reads and writes files your app stores
Progress notes	The agent records state for its next session
Subagents	Digest material in separate windows (module 4)

CURRENCY CHECK

Compaction and context editing run server-side and are in beta. The memory tool is client-side.

Compaction or memory?

COMPACTION

Keep this session going

- Summarises older turns
- Lossy: detail is gone
- Lives within the session

MEMORY

Carry facts across sessions

- Specific facts, written as files
- Your app stores and governs them
- Available next week

THE TIEBREAKER

To continue one long session, compact. To remember specific facts next time, write them to memory.

The trade-offs to remember

- **Clearing context invalidates the prompt cache** from that point onwards.
- **Compaction alone is not enough** for long-running agents: keep durable progress records.
- **Memory files are yours to govern:** scope them, set retention, block path tricks.
- **Clear only what is spent:** old tool output, not the findings drawn from it.

> ON THE EXAM
>
> Every context tool trades something. Know what each one throws away.

The output side of the budget

- **max_tokens is a hard cap** on output, thinking included.
- **Hit the cap and the reply stops mid-sentence**, or mid-JSON.
- **At high effort, leave generous room** for thinking plus the answer.
- **Output tokens cost about five times input** on current models.

> ON THE EXAM
>
> Raise max tokens when evidence shows truncation, not as a cost or quality fix.

Budgeting a request

01
Measure
Count the tokens per part.

02
Remove
What the task does not need.

03
Order
Stable first, question last.

04
Cache
The stable prefix.

05
Bound
max_tokens and a schema.

06
Monitor
Tokens per request, over time.

ON THE EXAM

You cannot optimise a budget you never measured. The token-counting endpoint costs nothing to call.

> Corvane is reviewing token tactics for Tallyhawk's support agent and sorting them by stage of the request.
>
> The team wants to know which tactics act on input before the prompt is assembled.

Which two tactics belong to input preparation? Choose two.

A Placing stable instructions first as a cacheable prefix

B Validating the reply against a schema

C Trimming retrieved help articles to the relevant passages

D Setting max tokens to fit the expected reply

E Summarising the earlier part of a long conversation

◆ PAUSE AND DECIDE

Which two tactics belong to input preparation? Choose two.

A Placing stable instructions first as a cacheable prefix

Prompt construction: it orders the prompt.

B Validating the reply against a schema

Output handling: it acts on the response.

C Trimming retrieved help articles to the relevant passages

Shrinks material before it enters the prompt.

D Setting max tokens to fit the expected reply

Output handling: it bounds the response.

E Summarising the earlier part of a long conversation

Compresses history before it enters the prompt.

THE SIGNAL

Input preparation shrinks material before the prompt; construction orders it; output handling bounds the reply.

> Tallyhawk's integration-diagnosis agent makes 90 or more tool calls on hard cases, many returning long log dumps. Late in a run its accuracy drops, and it repeats checks it has already done.
>
> The early findings matter for the final diagnosis.

Which change best addresses this?

A. Move the agent to a model with a larger context window

B. Restart the agent from scratch whenever the context gets full

C. Instruct the agent to write shorter replies

D. Clear old tool results automatically, and have the agent keep its findings in structured notes

◆ PAUSE AND DECIDE

Which change best addresses this?

A Move the agent to a model with a larger context window

Current models already hold a million tokens. Bigger is not better.

B Restart the agent from scratch whenever the context gets full

Throws away the early findings the diagnosis needs.

C Instruct the agent to write shorter replies

The bulk is tool output, not the agent's own words.

D Clear old tool results automatically, and have the agent keep its findings in structured notes

Removes spent log dumps while keeping what was learned.

THE SIGNAL

A window full of spent tool output: clear the output, keep the findings.

> Mereside's mortgage advisers use one assistant conversation per complex case, often over several days. Long conversations eventually approach the context limit, and quality slips.
>
> Advisers need the conversation to keep going, and key case facts must never be lost.

What should Corvane design?

A Start a fresh conversation each morning with no history

B Paste the full transcript into a new conversation each day

C Enable compaction so older turns are summarised, and record key case facts in the case system

D Raise max tokens so the conversation has more room

◆ PAUSE AND DECIDE

What should Corvane design?

A Start a fresh conversation each morning with no history

Continuity is the requirement. This drops it.

B Paste the full transcript into a new conversation each day

Same overloaded context, now copied by hand.

C Enable compaction so older turns are summarised, and record key case facts in the case system

Continuity from compaction; facts that must survive kept outside it.

D Raise max tokens so the conversation has more room

Max tokens limits output, not the window.

THE SIGNAL

Keep a long session going: compact. Facts that must never be lost: store them outside the window.

> Tallyhawk's support agent should remember each customer's set-up, such as their integrations and plan, across conversations weeks apart.
>
> Enterprise customers require that one customer's information is never visible to another.

Which design fits best?

A. A memory store the agent reads and writes per customer, scoped to that customer and governed by a retention policy

B. Keep every past conversation with the customer in the context window

C. Rely on compaction to carry the details from one conversation to the next

D. Add every customer's set-up notes to the shared system prompt

◆ PAUSE AND DECIDE

Which design fits best?

A A memory store the agent reads and writes per customer, scoped to that customer and governed by a retention policy

Specific facts across sessions, isolated per customer, under your control.

B Keep every past conversation with the customer in the context window

Ever-growing context, cost and context rot.

C Rely on compaction to carry the details from one conversation to the next

Compaction works within a session, not across weeks.

D Add every customer's set-up notes to the shared system prompt

Every customer's data in every conversation: a privacy breach.

THE SIGNAL

Specific facts across sessions, with isolation required: scoped memory you govern.

> Quillon's discharge summaries are returned as JSON. About 3% fail to parse because the object is cut off part way through.
>
> Logs show those responses stopped because they reached the max tokens limit. The calls run at high effort.

What should Corvane change?

A Retry failed calls until one completes

B Ask Claude to keep summaries shorter

C Strip the end of the JSON and repair it in code

D Raise max tokens to leave room for thinking and the full summary, and keep validating against the schema

◆ PAUSE AND DECIDE

What should Corvane change?

A Retry failed calls until one completes

Pays repeatedly for the same truncation.

B Ask Claude to keep summaries shorter

Risks dropping clinical content to fit a cap set too low.

C Strip the end of the JSON and repair it in code

Repairs the shape and silently loses the missing content.

D Raise max tokens to leave room for thinking and the full summary, and keep validating against the schema

The logs prove truncation; thinking shares the cap at high effort.

THE SIGNAL

Evidence of truncation at high effort: raise the cap. Without that evidence, raising it fixes nothing.

> A Corvane junior architect proposes sending all 1,800 of Mereside's policy documents with every agent-assist request, "because the model has a million-token window".
>
> The documents total several million tokens. Advisers ask about one or two policies per call.

What should the lead architect recommend?

A. Send as many documents as fit, with prompt caching to reduce the cost

B. Retrieve only the passages relevant to each question, and keep the stable instructions cached

C. Summarise every document into one long digest and send it every time

D. Split the documents across three parallel calls and merge the answers

◆ PAUSE AND DECIDE

What should the lead architect recommend?

A. Send as many documents as fit, with prompt caching to reduce the cost

Cheaper, but still mostly irrelevant tokens competing for attention.

B. Retrieve only the passages relevant to each question, and keep the stable instructions cached

The smallest high-signal context for each question.

C. Summarise every document into one long digest and send it every time

Loses the exact policy wording advisers must rely on.

D. Split the documents across three parallel calls and merge the answers

Triples the cost to read mostly irrelevant material.

THE SIGNAL

Capacity is not strategy. A large, varied corpus and narrow questions: retrieve what is relevant.

In five lines

- **A big window is capacity, not strategy.** Aim for the smallest high-signal context.
- **Prepare, construct, handle output:** shrink, order, then bound.
- **Extract for evidence; summarise for continuity.**
- **Compaction continues a session; memory carries facts across sessions;** clear only what is spent.
- **max tokens caps output, thinking included.** Raise it on evidence of truncation.

COMING UP

Next: prompt reuse. Caching, modular prompts and Skills.